

At the end of the day, a driver will expect to know what his earnings are—the cash collected by him and the earnings credited by the cashless transactions.

The driver should be able to provide feedback about the customer as well, although one expects all customers to behave in a civil manner with the driver.

App for the customers

A client application would obviously be similar to the ones that many of us have used. The functionality has to be similar, though the user interfaces could be better. A local taxi aggregator may not be as interested in pushing other products and services on the main page.

The app will need to provide information about the availability of taxis, as well as the approximate time and cost for a ride. Clients should be able to request for a taxi to pick them up at the current or another location. There should be an option to schedule a ride.

A client should be able to contact the driver if need be, especially if the driver appears to be headed towards an incorrect location for the pick-up.

At the end of the trip, the client should be able to pay by his or her preferred method and provide feedback in case needed. A desirable capability would be that a client should have control over the amount of customer history that the taxi aggregator can keep.

Optimisation goals

The goals for a local taxi union will be very different from the current model. The taxi union will obviously want to maximise the earnings of its members. However, that does not necessarily conflict with the customers' needs. Both would want to wait for as little time as possible. A taxi driver will not like his return journey to be in an empty vehicle.

A local taxi union will also need to use Big Data and statistical modelling to optimise the movement of taxis. The best way to do this would be by finding ways to optimise the fare a client pays.

There may still be time- and traffic-dependent discounts in fares for travelling in certain directions. In case of a shortage of taxis, the shared taxi options may be made more attractive.

Each taxi aggregator may add the functionality and models appropriate for its area, which may then be adapted and extended by other communities.

Each of these could be interesting and useful projects for computer engineering students. So, if some students start on it, who knows, some taxi union somewhere in the world may take it up from there. 

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